ALA-TOO INTERNATIONAL UNIVERSITY

BUSINESS ADMINISTRATION SOFTWARE

Spring Semester 2025-2026

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What gets measured gets managed.

– Peter Drucker

WHAT WAS PROMISED LAST WEEK

ERP/CRM DATA → DASHBOARDS

Last lecture ended with a promise:
How data from ERP and CRM systems becomes KPIs, reports and dashboards used by managers

Today's lesson delivers that promise and explains how dashboards can be correct or dangerously misleading.

DESIGN KPIS THAT MATTER

TODAY'S TARGET SKILL

By the end of today, you should be able to:

Explain what a KPI is (formally and practically)

Explain the difference between a metric, KPI, report and dashboard

Explain why "**nice dashboards**" can still lead to wrong decisions

Build simple KPI + decision logic (as if you are building the backend of a dashboard)

WHY MANAGERS NEED DASHBOARDS

HUMANS CAN’T READ LOGS

ERP/CRM systems store thousands of records per day: orders, invoices, leads, calls, deliveries.

Managers cannot read logs like IT people. They need answers fast:

Are we OK?

What is getting worse?

What must be fixed today, not next month?

Dashboards exist to compress complexity into decisions.

GAME HUD VS RAW GAME DATA

ANALOGY

In a game, the computer calculates hundreds of variables every second.
The player does not see all of that. The player sees a HUD:

health bar
ammo
minimap
mission objective

A dashboard is a business HUD.
It shows only what matters to decide the next move.

DIGITAL TRANSFORMATION

FROM MANUAL TO CONTROLLED

Digital transformation means moving from "**people-based operations**" to "**system-based operations**"

It is not about buying computers, it is about changing how work happens:

work is standardized

steps are recorded

rules are enforced

performance becomes measurable

If work is not recorded digitally, dashboards cannot be trusted.

DIGITALIZATION VS AUTOMATION

WHOLE PROCESS VS ONE STEP

Automation is when one step becomes faster.
Example: sending an invoice email automatically

Digitalization is when the entire workflow becomes controlled.
Example: invoice created → approved → issued → payment tracked → overdue flagged → report updated.

Dashboards rely on digitalization because dashboards need data from the full workflow, not only one task.

SPREADSHEETS ARE STAGE 1

EXCEL AS EARLY SYSTEM

Many businesses start with Excel/Google Sheets for:

inventory lists

simple budgets

sales tracking

customer contacts

This is normal. Spreadsheets are quick, flexible, and cheap.
But they are not designed for scale, control, or single truth.

WHY EXCEL BREAKS

MULTIPLE VERSIONS AND ERRORS

Example: group project files → Final, Final2, Final_REAL, Final_FINAL.
ONLY THE LAST EDITOR kinda knows which one is correct.

Same in business:
multiple sheet copies
manual edits
formulas changed accidentally
no audit trail

When a company grows, Excel becomes a risk, not a tool.

ACCOUNTING SYSTEMS

MONEY BECOMES DATA

Accounting software turns financial events into structured records:

Invoices (sum of issued sales bills for goods or services)
Payments (total money received from customers)
Expenses (sum of all recorded business costs)
Taxes (percentage of taxable income owed to the government)
Profit/Loss reports (revenue minus expenses for a specific period

Accounting systems are the "**official memory**" of money.
Dashboards need this because financial KPIs must be based on verified records, not guesses.

ACCOUNTING FEEDS DASHBOARDS

FINANCIAL TRUTH SOURCE

Sales may say: We sold a lot.

Accounting answers: How much cash came in? What was the profit?

Examples of dashboard KPIs that require accounting data:

Gross profit (revenue minus cost of goods sold)

Net income (final profit after all expenses)

Cash collected vs invoiced (money received vs invoices issued)

Expense trends (how company costs change over time)

If accounting data is delayed or incomplete, dashboards look nice but are wrong.

ERP SYSTEMS

OPERATIONS + FINANCE IN ONE

ERP systems integrate operations with finance and standardize processes across departments. ERP makes operations measurable because it records events like:

Inventory movement (quantity of items added to or removed from stock)
Procurement orders (total value or number of purchase orders issued to suppliers)
Warehouse actions (count of picking, packing, and storage operations performed)
Production status(number of units produced vs planned production target)
Shipping and delivery (orders shipped or delivered within a time period)

ERP is the foundation for operational KPIs.

CRM SYSTEMS

CUSTOMERS AND PIPELINE DATA

CRM provides demand signals and revenue forecasting data used in dashboards. CRM systems manage customer-related activity:

Leads and opportunities (number of potential and qualified customers)
Pipeline stages (current step of a deal in the sales process)
Deal values and probabilities (deal revenue adjusted by chance of closing)
Interaction history (calls/emails/meetings) (record of customer communications)

DASHBOARDS DEPEND ON SYSTEMS

KEY POINT

Dashboards **DO NOT** create accuracy.
They display what systems record.

If the system data is messy, the dashboard becomes a confident lie.
Good dashboards start with good processes and clean data sources.

RECEIPTS VS MEMORY

ANALOGY

If you try to remember spending from memory, you will be wrong.
If you check bank statements and receipts, you can prove the truth.

ERP/CRM/Accounting are receipts.
Dashboards are a summary of receipts.

KPI CORE: DEFINITION + STRUCTURE

WHAT IS A KPI

FORMAL DEFINITION

WHAT IS A KPI

FORMAL DEFINITION

A KPI (Key Performance Indicator) is a measurable value used to evaluate how effectively a business objective is being achieved.

A KPI is tied to:

Goal (the business objective being pursued)
Method of measurement (how performance is calculated)
Target / threshold (the value that defines success or risk)
Decision action (the response triggered when limits are reached)

KPIs are designed to guide decisions, not to decorate reports.

KPI VS METRIC

NOT ALL NUMBERS MATTER

A metric is any measurement.

Example: number of calls made, number of website visits.

A KPI is a metric that directly supports an objective and triggers action.

Example: conversion rate in the sales funnel because it predicts revenue.

If the number does not affect decisions, it should not be a KPI.

KPI STRUCTURE

WHAT MAKES IT VALID

A KPI must:

Link to an objective (what goal does it support?)

Have ownership (who is accountable?)

Have measurable limits (what is OK vs NOT OK?)

Trigger action (what do we do when it fails?)

Without ownership and action, KPIs become “statistics” with no business value.

KPI ANALOGY

EXAM SCORE WITH THRESHOLD

Objective: pass the course

KPI: exam score

Threshold: 50% to pass

If score < 50% → action is required (study plan, консультация, retake preparation)

This is exactly how business KPIs work: threshold + response

LEADING KPIs

EARLY WARNING SIGNALS

Leading KPIs predict what will happen later.
They show problems early, before revenue or profit changes.

Examples:

conversion rate dropping predicts future revenue decline

sales cycle length increasing predicts delayed cash flow

customer engagement falling predicts churn risk

Leading KPIs allow intervention before damage becomes permanent.

LAGGING KPIs

CONFIRMING RESULTS

Lagging KPIs measure outcomes after they occur.
They confirm how the business performed in the past.

Examples:

revenue (total money from sales)
gross profit (revenue minus direct costs)
net income (profit after all costs and taxes)
churn rate (customers who already left)

Lagging KPIs are important for truth, but they often arrive too late to prevent problems.

HEALTHY DASHBOARD MIX

LEADING + LAGGING TOGETHER

A healthy dashboard includes:

Leading KPIs to predict

Lagging KPIs to confirm

Example:
If you only look at final exam results, you find out too late.
If you also track weekly quizzes, attendance, and assignments, you can fix issues early.

Same idea in business: early signals + final outcomes

THRESHOLDS

WHEN KPI BECOMES CONTROL

KPIs must include thresholds to trigger action.
Without limits, dashboards become passive.

Examples:

late deliveries > 3% → investigate logistics

margin < 15% → review costs and pricing

conversion < 10% → fix sales qualification

Thresholds convert measurement into control.

KPI LIFECYCLE

KPIS MUST EVOLVE

KPIs must change when strategy changes.

Example:

Early stage: focus on growth → track acquisition and conversion

Later stage: focus on profitability → track margin and cost efficiency

Static KPIs mislead teams into optimizing outdated goals.
Companies must review KPIs regularly to stay aligned with reality.

KPI GAMING

WRONG KPI → WRONG BEHAVIOR

If performance is measured only by “calls per day,” calls increase, but quality decreases.

People optimize what you measure.

Better KPI design focuses on outcomes:

qualified opportunities (sales prospects that meet buying criteria)

conversion rate (percentage of prospects that become customers)

revenue per salesperson (average sales generated by each salesperson)

customer satisfaction (customer rating of service or product quality)

KPIs shape behavior like rules shape a game.

DASHBOARDS: SAFE DESIGN, TRUST, CONSISTENCY

REPORTS VS DASHBOARDS

PAST VS INTERVENTION

Reports explain history:
“What happened last month?”
They are static and used for documentation.

Dashboards enable intervention:
“What is happening now?”
They are dynamic and used to trigger action.

Report = history book.
Dashboard = control panel.

DASHBOARD DESIGN RULE

ONE ROLE, ONE PURPOSE

Dashboards must be role-specific.

Example:

Warehouse dashboard: fulfillment time, inventory aging

Sales manager dashboard: conversion, pipeline velocity

CFO dashboard: margin, cash flow

If you build one dashboard for everyone, it becomes clutter and nobody trusts it.

WHY NUMBERS DISAGREE

MULTIPLE DEFINITIONS

Sales revenue ≠ Finance revenue when definitions differ.

Example:

CRM counts revenue when a deal is “won”

Accounting counts revenue when invoice is issued/paid

Both might be correct by their own rules, but management needs one consistent truth.

IT RESPONSIBILITY

ONE VERSION OF TRUTH

IT must enforce:

standard metric definitions

consistent calculation logic

shared time periods and currency rules

documented data sources (ERP/CRM/Accounting)

Without this governance, dashboards become political arguments instead of decision tools.

TRENDS OVER SNAPSHOTS

DIRECTION IS THE SIGNAL

A single number is a snapshot.

A trend shows direction.

Example:

Revenue: 1.2M → 1.1M → 1.0M

Snapshot “1.0M” can look fine, but the trend is a warning.

Trends are what drive strategy and planning.

MISLEADING DASHBOARD

WHEN IT LOOKS “OK”

Revenue appears stable.
Lead volume is increasing.

But:

conversion is falling

margins are shrinking

This is dangerous because activity is rising, but efficiency is collapsing.
The business can look stable while profitability slowly dies.

BUSINESS CONSEQUENCE

WRONG DECISION IMPACT

If management increases marketing based on “more leads”:

marketing spend rises

sales team workload rises

conversion stays weak

costs increase

Profit shrinks even if revenue stays stable.

A dashboard interpretation failure becomes a financial failure.

GAME: TWO CONNECTED CODING TASKS (CLEAR + GRADABLE)

CODING GAME

BUILD A MINI DASHBOARD ENGINE

You will build the logic behind a dashboard, not just formulas.

Task 1 computes KPI values from pipeline + costs.

Task 2 takes Task 1 output and produces exactly one management decision.

This models how analytics systems work: data → KPI → action.

TASK 1 INPUT

PIPELINE + COST DATA

Given input values (single scenario):

Leads = 200

Opportunity rate = 30%

Close rate = 20%

Deal value = 2000

Marketing cost = 6000

Operational cost = 9000

Goal: convert this raw input into KPI metrics a dashboard would display.

TASK 1 CALCULATIONS

WHAT MUST BE COMPUTED

Compute step-by-step:

Opportunities = Leads × Opportunity rate

Closed deals = Opportunities × Close rate

Revenue = Closed deals × Deal value

Profit = Revenue − (Marketing cost + Operational cost)

CAC = Marketing cost ÷ Closed deals

Output must follow exact format:

REVENUE=____ PROFIT=____ CAC=____

TASK 1 ANSWER

VALIDATION NUMBERS

$$\text{Opportunities} = 200 \times 0.30 = 60$$

$$\text{Closed deals} = 60 \times 0.20 = 12$$

$$\text{Revenue} = 12 \times 2000 = 24000$$

$$\text{Profit} = 24000 - (6000 + 9000) = 9000$$

$$\text{CAC} = 6000 \div 12 = 500$$

Expected output:

REVENUE=24000 PROFIT=9000 CAC=500

TASK 2 RULES

FROM KPI TO DECISION

Using Task 1 output, print exactly one decision:

Rules:

If PROFIT < 0 → CUT COSTS

Else if CAC > 700 → IMPROVE CONVERSION

Else → SCALE MARKETING

This simulates how dashboards lead to actions.

TASK 2 ANSWER

ONE CORRECT DECISION

From Task 1:

PROFIT = 9000 (positive)

CAC = 500 (not high)

So decision must be:
SCALE MARKETING

One correct output = easy grading.

NEXT WEEK PREVIEW

PRACTICE DASHBOARD DESIGN

How to …

Choose KPIs for a role (Sales / Operations / Finance)

Set thresholds and actions

Prevent “multiple truths” with a KPI definition table

Build a small dashboard model (even on paper) before any software